

EL — Elements of Life

OCR B (Salters) H433 storyline **EL**. Where the elements come from: atomic structure and spectra, isotopes and mass spectrometry, the periodic table and Group 2, bonding and the shapes of molecules, amount of substance and titrations.

1. Atomic Structure

The Atom

Atom: The smallest particle of an element that can take part in a chemical reaction. Atoms consist of a small, dense, positively charged nucleus containing protons and neutrons, surrounded by electrons in energy levels (shells).

Particle	Relative Mass	Relative Charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	$1/1836 \approx 0$	-1	Shells / orbitals

The **atomic number (Z)** is the number of protons in the nucleus and defines the element. The **mass number (A)** is the total number of protons and neutrons. For a neutral atom the number of electrons equals the number of protons.

Isotopes

Isotopes: Atoms of the same element (same number of protons) with different numbers of neutrons and therefore different mass numbers. Isotopes have identical chemical properties because they have

the same electron configuration, but different physical properties (e.g. density, rate of diffusion).

Examples: carbon-12 (^{12}C , 6p + 6n) and carbon-13 (^{13}C , 6p + 7n). The existence of isotopes means most elements do not have whole-number relative atomic masses.

Mass Spectrometry

A mass spectrometer measures the relative masses and abundances of isotopes or molecules. The basic steps are:

1. **Ionisation:** The sample is vaporised and bombarded with high-energy electrons to form positive ions (electron impact ionisation) — an electron is knocked off each atom/molecule.
2. **Acceleration:** Ions are accelerated by an electric field so they all have the same kinetic energy.
3. **Deflection:** Ions pass through a magnetic field. Lighter ions and ions with higher charge are deflected more.
4. **Detection:** Ions reach the detector, producing a current proportional to their abundance.

The mass spectrum shows peaks at each m/z value. The relative atomic mass (A_r) of an element is calculated from the mass spectrum using:

$$A_r = \Sigma(\text{isotope mass} \times \% \text{ abundance}) / 100$$

Exam Tip: In mass spectra of molecules, the peak with the highest m/z value (ignoring the $M+1$ peak) is the molecular ion peak (M^+), giving the relative molecular mass. Fragmentation patterns help identify functional groups and structural features.

Worked Example — Calculating A_r :

Chlorine has two isotopes: ^{35}Cl (abundance 75.8%) and ^{37}Cl (abundance 24.2%).

$$A_r = (35 \times 75.8 + 37 \times 24.2) / 100 = (2653.0 + 895.4) / 100 = \mathbf{35.5}$$

Electron Configuration

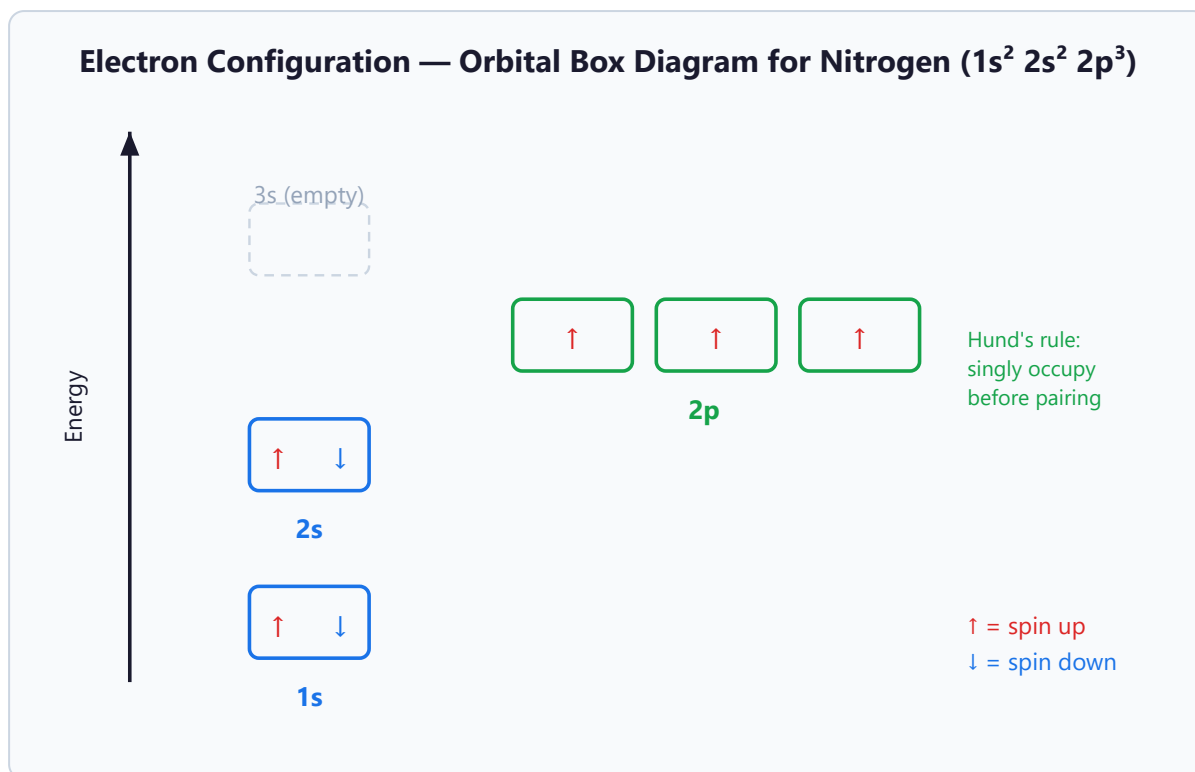
Electrons occupy atomic orbitals. An orbital is a region of space around the nucleus where there is a high probability of finding an electron. Each orbital holds a maximum of two electrons with opposite spins (Pauli exclusion principle).

Sub-shell	Number of orbitals	Max electrons	Shape
s	1	2	Spherical
p	3	6	Dumbbell
d	5	10	Complex / cloverleaf
f	7	14	Complex

The order of filling follows the Aufbau principle: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p ... Note that 4s fills before 3d. When writing electron configurations for ions, remove electrons from the highest principal energy level first (e.g. Fe: [Ar] 3d⁶ 4s², but Fe²⁺: [Ar] 3d⁶).

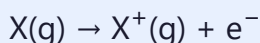
Hund's rule: Electrons occupy orbitals of the same sub-shell singly before pairing up, and with parallel spins. This minimises electron-electron repulsion.

Chromium and copper are exceptions: Cr is [Ar] 3d⁵ 4s¹ (not 3d⁴ 4s²) and Cu is [Ar] 3d¹⁰ 4s¹ (not 3d⁹ 4s²) because half-filled and fully-filled d sub-shells have extra stability due to symmetrical electron distribution and favourable exchange energy.



Ionisation Energies

First ionisation energy: The energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous $1+$ ions.



Ionisation energy depends on: nuclear charge, atomic radius, and shielding (screening) by inner electrons.

Trends:

- **Across a period:** Generally increases because nuclear charge increases while shielding remains roughly constant, so the outer electron is held more tightly.
- **Down a group:** Decreases because the atomic radius increases and shielding increases, outweighing the increase in nuclear charge.

There are two anomalies across Period 3: a dip from Mg to Al (the 3p electron in Al is easier to remove than a 3s electron in Mg because the 3p sub-shell is higher in energy and slightly shielded by 3s) and a dip from P to S (the paired 3p electron in S experiences extra repulsion, making it easier to remove).

Exam Tip: Successive ionisation energies can reveal the number of electrons in each shell. A large jump between successive values indicates that the next electron is being removed from a shell closer to the nucleus (less shielded, more tightly held).

2. Amount of Substance

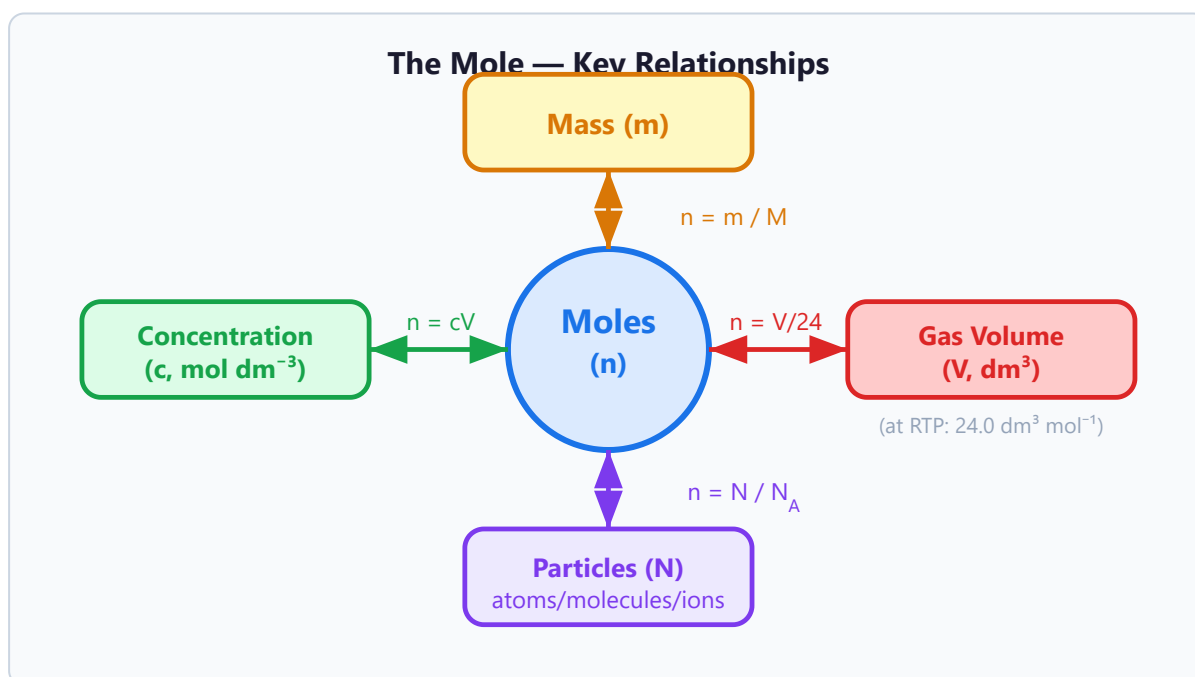
The Mole and Avogadro's Constant

Mole: The amount of substance that contains 6.022×10^{23} particles (atoms, molecules, ions, etc.). This number is Avogadro's constant (N_A).

Molar mass (M): The mass of one mole of a substance, in g mol^{-1} . Numerically equal to the relative formula mass.

$$n = m / M$$

where n is the number of moles, m is the mass in grams, and M is the molar mass in g mol^{-1} .



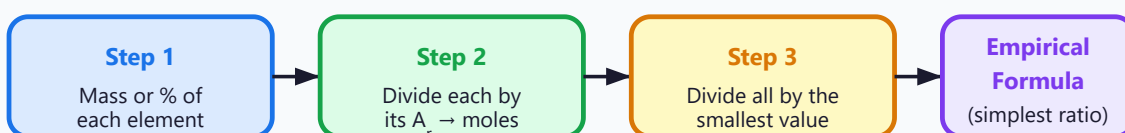
Empirical and Molecular Formulae

Empirical formula: The simplest whole-number ratio of atoms of each element in a compound.

Molecular formula: The actual number of atoms of each element in one molecule of a compound.

To find the empirical formula from percentage composition: divide each percentage by the element's A_r , then divide all results by the smallest value to get the simplest whole-number ratio. The molecular formula is a whole-number multiple of the empirical formula: molecular formula = (empirical formula) $\times n$, where $n = M_r / \text{empirical formula mass}$.

Finding Empirical & Molecular Formulae



Worked Example: Compound with 40.0% C, 6.7% H, 53.3% O

Element	% / A_r	Moles	Ratio
C	40.0 / 12.0	3.33	1
H	6.7 / 1.0	6.70	2
O	53.3 / 16.0	3.33	1

Empirical formula = CH_2O

If $M_r = 60$: $n = 60/30 = 2$, so molecular formula = $\text{C}_2\text{H}_4\text{O}_2$ (ethanoic acid)

Molar Gas Volume

At room temperature and pressure (RTP, approximately 25 °C and 1 atm), one mole of any gas occupies approximately 24.0 dm³ (24,000 cm³).

$$n = V / V_m$$

where V is the volume in dm³ and $V_m = 24.0 \text{ dm}^3 \text{ mol}^{-1}$ at RTP.

The ideal gas equation provides a more general relationship:

$$pV = nRT$$

where p is pressure in Pa, V is volume in m^3 , n is moles, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, and T is temperature in K.

Concentrations of Solutions

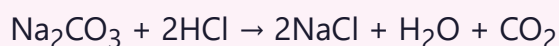
$$c = n / V$$

where c is concentration in mol dm^{-3} , n is moles of solute, and V is volume of solution in dm^3 . To convert cm^3 to dm^3 , divide by 1000.

Exam Tip: Always check your units. A very common error is using volume in cm^3 instead of dm^3 in the concentration formula. Remember: $1 \text{ dm}^3 = 1000 \text{ cm}^3$. For $pV = nRT$, pressure must be in Pa, volume in m^3 , and temperature in K.

Worked Example — Titration Calculation:

25.0 cm^3 of Na_2CO_3 solution of unknown concentration was titrated with $0.100 \text{ mol dm}^{-3}$ HCl. The mean titre was 27.60 cm^3 .



$$n(\text{HCl}) = 0.100 \times 27.60/1000 = 2.76 \times 10^{-3} \text{ mol}$$

$$n(\text{Na}_2\text{CO}_3) = 2.76 \times 10^{-3} / 2 = 1.38 \times 10^{-3} \text{ mol (1:2 ratio)}$$

$$[\text{Na}_2\text{CO}_3] = 1.38 \times 10^{-3} / 0.0250 = \mathbf{0.0552 \text{ mol dm}^{-3}}$$

Balanced Equations and Reacting Masses

A balanced chemical equation shows the molar ratios of reactants and products. Use these ratios to calculate reacting masses, volumes of gases, and volumes of solutions. The general approach:

1. Write the balanced equation.
2. Calculate moles of the known substance.
3. Use the molar ratio to find moles of the required substance.

4. Convert moles to the desired quantity (mass, volume, concentration).

Percentage Yield and Atom Economy

$$\text{Percentage yield} = (\text{actual yield} / \text{theoretical yield}) \times 100\%$$

$$\text{Atom economy} = (\text{M}_r \text{ of desired product} / \Sigma \text{M}_r \text{ of all products}) \times 100\%$$

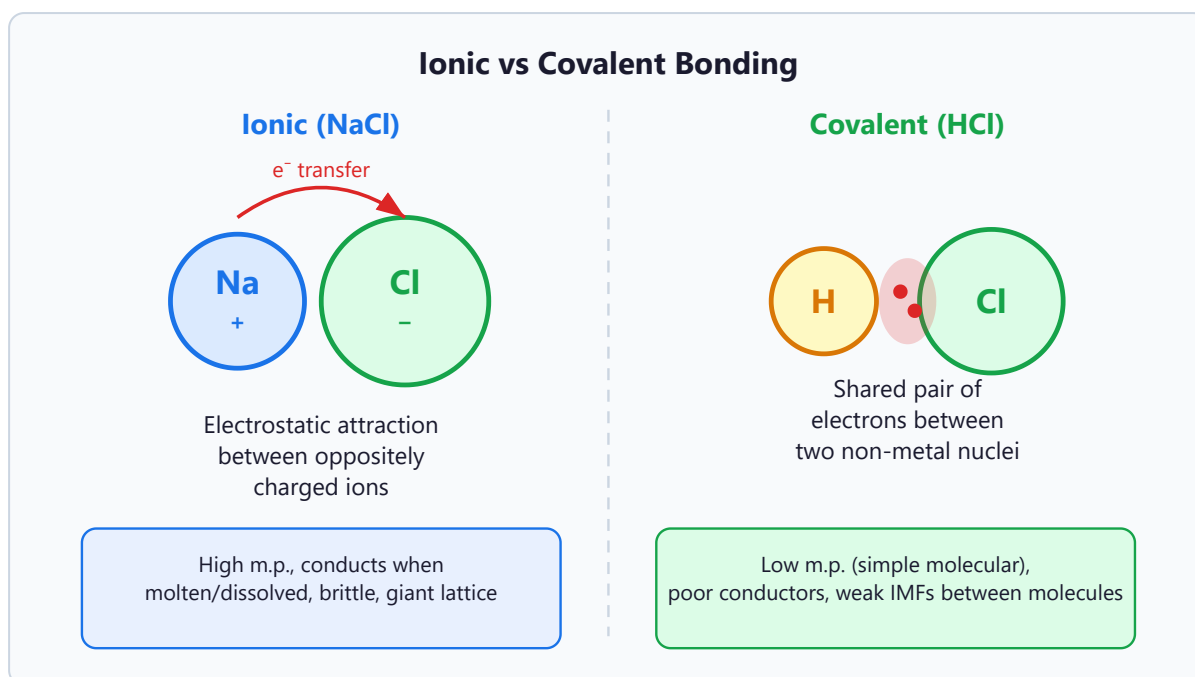
A high atom economy means less waste. Addition reactions have 100% atom economy. Substitution and elimination reactions have lower atom economies because by-products are formed.

3. Bonding and Structure

Ionic Bonding

Ionic bonding: The electrostatic attraction between oppositely charged ions, formed by the transfer of electrons from a metal to a non-metal. Metals lose electrons to form cations; non-metals gain electrons to form anions.

Ionic compounds form giant ionic lattices (e.g. NaCl). Properties: high melting points (strong electrostatic forces throughout the lattice), conduct electricity when molten or dissolved (ions are free to move), brittle (displacement of layers causes like charges to align and repel), and dissolve in polar solvents such as water.



Covalent Bonding

Covalent bond: A shared pair of electrons between two atoms, where both nuclei are attracted to the shared electron density. This occurs between non-metal atoms.

Dative (coordinate) bond: A covalent bond in which both electrons in the shared pair come from the same atom (the donor). Represented by an arrow from donor to acceptor. Example: NH_4^+ (the lone pair on nitrogen is donated to H^+).

A **single bond** consists of one shared pair (σ bond). A **double bond** consists of two shared pairs (one σ + one π bond). A **triple bond** consists of three shared pairs (one σ + two π bonds).

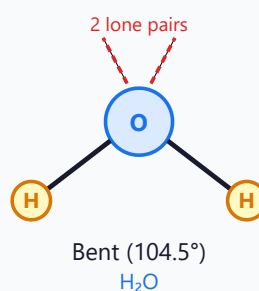
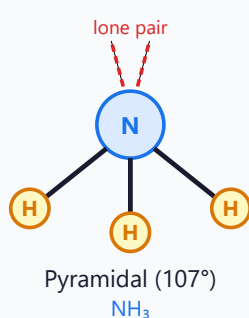
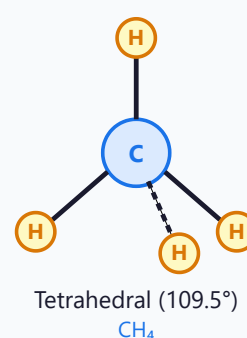
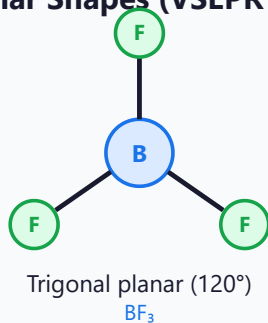
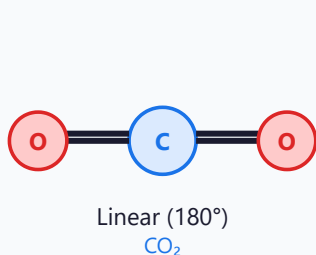
Shapes of Molecules — VSEPR Theory

VSEPR (Valence Shell Electron Pair Repulsion) theory: Electron pairs around a central atom arrange themselves to be as far apart as possible to minimise repulsion. Lone pairs repel more than bonding pairs: lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair.

Electron pairs	Bonding pairs	Lone pairs	Shape	Bond angle	Example
----------------	---------------	------------	-------	------------	---------

2	2	0	Linear	180°	CO ₂ , BeCl ₂
3	3	0	Trigonal planar	120°	BF ₃ , AlCl ₃
3	2	1	Bent / V-shape	~117°	SO ₂
4	4	0	Tetrahedral	109.5°	CH ₄ , NH ₄ ⁺
4	3	1	Trigonal pyramidal	107°	NH ₃
4	2	2	Bent / V-shape	104.5°	H ₂ O
5	5	0	Trigonal bipyramidal	90° & 120°	PCl ₅
6	6	0	Octahedral	90°	SF ₆

Molecular Shapes (VSEPR Theory)



Lone pairs reduce bond angles: LP-LP > LP-BP > BP-BP repulsion

Electronegativity and Polarity

Electronegativity: The ability of an atom to attract the bonding pair of electrons in a covalent bond towards itself. Measured on the Pauling scale. Fluorine is the most electronegative element (4.0).

Trends: Electronegativity increases across a period (greater nuclear charge, smaller atomic radius) and decreases down a group (greater atomic radius, more shielding).

A **polar bond** exists when there is a difference in electronegativity between two bonded atoms. The more electronegative atom carries a partial negative charge (δ^-) and the less electronegative atom carries a partial positive charge (δ^+). A molecule is polar overall only if the individual bond dipoles do not cancel (i.e. the molecule is not symmetrical). For example, CO_2 has polar bonds but is a non-polar molecule (linear, dipoles cancel), whereas H_2O is a polar molecule (bent shape, dipoles do not cancel).

Metallic Bonding

Metallic bonding: The electrostatic attraction between a lattice of positive metal ions and a sea of delocalised electrons. The outer-shell electrons are free to move throughout the structure.

Properties of metals: high melting points (strong metallic bonds), good electrical conductors (delocalised electrons carry charge), good thermal conductors (delocalised electrons transfer kinetic energy), malleable and ductile (layers of ions can slide over each other without breaking the bonding).

Intermolecular Forces

Intermolecular forces are weak forces between molecules. They determine physical properties such as boiling point and solubility.

Type	Strength	Origin	Present in
London (dispersion) forces	Weakest	Temporary instantaneous dipoles caused by random electron movement induce dipoles in neighbouring molecules	All molecules
Permanent dipole–dipole	Moderate	Attraction between the δ^+ end of one polar molecule and the δ^- end of another	Polar molecules

Hydrogen bonding	Strongest intermolecular force	A special, strong dipole–dipole interaction between a lone pair on N, O, or F and a hydrogen atom bonded to N, O, or F	NH ₃ , H ₂ O, HF, alcohols, carboxylic acids
------------------	--------------------------------	--	--

London forces increase with the number of electrons (and therefore molecular size) because larger electron clouds are more easily polarised, creating stronger temporary dipoles.

Exam Tip: When comparing boiling points, always identify the strongest type of intermolecular force present. Hydrogen bonding explains why water ($M_r = 18$) has a much higher boiling point than methane ($M_r = 16$) despite similar molecular masses.

Giant Structures

Giant ionic lattices (e.g. NaCl): Strong electrostatic forces between ions in all directions → high melting points, conduct when molten/dissolved.

Giant covalent (macromolecular) structures (e.g. diamond, silicon dioxide, graphite): Many strong covalent bonds throughout the structure → very high melting points, generally insoluble, generally do not conduct electricity. Exception: graphite has delocalised electrons in the region between the layers, free to move *along* them, so it conducts electricity parallel to the layers (it is a poor conductor perpendicular to them); the weak London forces between layers make it soft and slippery.

Simple molecular structures (e.g. I₂, H₂O, CO₂): Strong covalent bonds within molecules but weak intermolecular forces between molecules → low melting/boiling points, do not conduct electricity (no charged particles free to move).

Exam Tip: When explaining high melting points, always state the type of bonding/force that must be overcome and describe it as strong. For simple molecular substances, make clear it is the intermolecular forces (not the covalent bonds) that break during melting and boiling. Never say "bonds are broken" when discussing boiling — use "intermolecular forces are overcome."

4. Nuclear Fusion and the Origin of the Elements

The Salters storyline opens with the question of where the elements came from. The Big Bang produced essentially only hydrogen and helium (with a trace of lithium); every heavier nucleus in your body was assembled inside a star.

Nuclear fusion: A reaction in which two lighter nuclei join together to give a heavier nucleus.

Because all nuclei are positively charged, they repel one another strongly, so fusion only happens at the very high temperatures and pressures found in stellar cores — the nuclei need enormous kinetic energy to get close enough for the short-range strong nuclear force to take over.

Fusion is a *nuclear* change, not a chemical one. The identity of the element changes because the number of protons changes, and the energy released per gram is millions of times larger than for any chemical reaction.

Writing Nuclear Equations

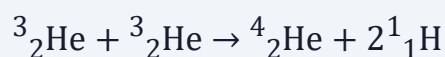
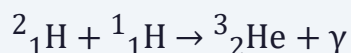
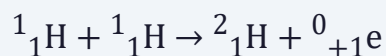
Each species is written as A_ZX , where A is the mass number (protons + neutrons) and Z is the atomic number (protons). A nuclear equation balances when:

- the sum of the mass numbers is the same on both sides, and
- the sum of the atomic numbers (charges) is the same on both sides.

Particle	Symbol	Mass number	Atomic number
Proton (hydrogen nucleus)	${}^1_1\text{H}$	1	+1
Neutron	${}^1_0\text{n}$	1	0
Alpha particle (helium nucleus)	${}^4_2\text{He}$	4	+2
Positron	${}^0_{+1}\text{e}$	0	+1
Gamma photon	γ	0	0

Hydrogen Burning

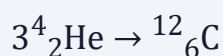
In a main-sequence star such as the Sun, four hydrogen nuclei are converted to one helium-4 nucleus through a sequence of steps:



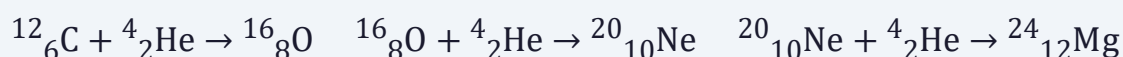
The overall change is $4{}^1_1\text{H} \rightarrow {}^4_2\text{He} + 2{}^0_{+1}\text{e}$. Mass numbers: $4 = 4$. Atomic numbers: $4 = 2 + 2(+1) = 4$.

Helium Burning and Beyond

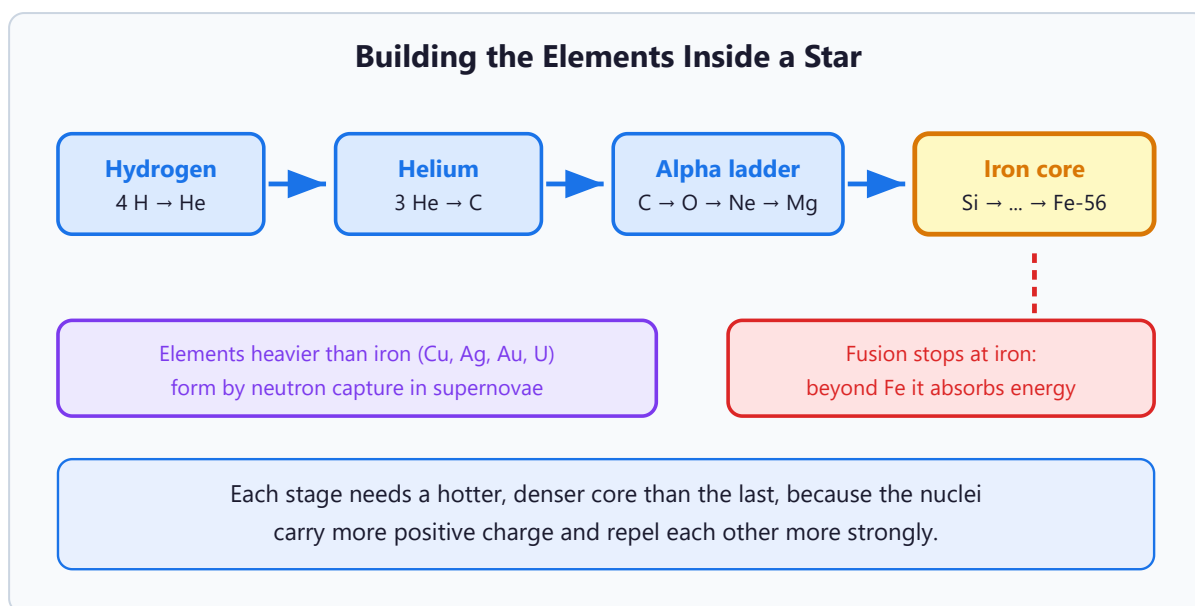
When the hydrogen in the core is exhausted, the star contracts, the core temperature rises and helium nuclei begin to fuse. The **triple-alpha process** builds carbon:



Successive additions of alpha particles then build the familiar "alpha ladder" of abundant even-numbered elements:



This explains why carbon, oxygen, neon, magnesium, silicon, sulfur and calcium — all multiples of four mass units — are far more abundant in the universe than their odd-numbered neighbours.



Why Fusion Stops at Iron

Binding energy per nucleon rises from hydrogen up to a maximum around iron and nickel, then falls again. Fusing nuclei *lighter* than iron therefore releases energy, but fusing nuclei *heavier* than iron would absorb it. Once a massive star has built an iron-56 core it can no longer generate energy by fusion, the core collapses and the star explodes as a supernova. Elements beyond iron — copper, silver, gold, uranium — are made by nuclei capturing neutrons in and around such explosions, not by ordinary fusion.

Worked Example — Balancing a Nuclear Equation:

Complete: $^{14}_7\text{N} + ^4_2\text{He} \rightarrow ^{17}_8\text{O} + ?$

Mass numbers: $14 + 4 = 18$ on the left; 17 on the right so far, so the missing particle has $A = 18 - 17 = 1$.

Atomic numbers: $7 + 2 = 9$ on the left; 8 on the right so far, so the missing particle has $Z = 9 - 8 = 1$.

$A = 1$ and $Z = 1$ is a proton, ^1_1H . The completed equation is $^{14}_7\text{N} + ^4_2\text{He} \rightarrow ^{17}_8\text{O} + ^1_1\text{H}$.

Exam Tip: Fusion questions are almost always "balance the nuclear equation" questions, and they are free marks if you are systematic: total the top numbers, total the bottom numbers, and read off what is missing. Do not confuse fusion (light nuclei joining, needs very high temperature and pressure)

with fission (a heavy nucleus splitting). And note that a positron has mass number 0 but atomic number +1 — forgetting it is the commonest way to get the charge balance wrong.

5. Group 2 Chemistry

Group 2 — the alkaline earth metals — enters the storyline through the salts found in sea water, in bath salts, in barium meals and in fertilisers. H433 requires magnesium, calcium, strontium and barium.

Trends Down the Group

Each element has the outer configuration ns^2 and forms a $2+$ ion by losing both outer electrons. Going down the group:

- atomic radius increases (an extra occupied shell each time);
- shielding by inner shells increases;
- so the outer electrons are attracted less strongly and the first (and second) ionisation enthalpies **decrease**;
- so the elements lose electrons more easily and **reactivity increases** down the group.

Exam Tip: Nuclear charge does increase down the group, so you must say why it loses: the increase in distance and shielding outweighs the increase in nuclear charge. An answer that only says "more shells" or only says "more protons" will not get full marks.

Reaction with Water

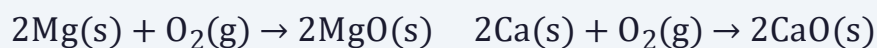
Element	With cold water	Equation
Mg	Extremely slow; reacts readily with steam instead, burning with a bright white flame	$\text{Mg} + \text{H}_2\text{O(g)} \rightarrow \text{MgO} + \text{H}_2$
Ca	Steady effervescence; a cloudy white suspension of sparingly soluble Ca(OH)_2	$\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$

Sr	Faster effervescence	$\text{Sr} + 2\text{H}_2\text{O} \rightarrow \text{Sr}(\text{OH})_2 + \text{H}_2$
Ba	Vigorous effervescence; the hydroxide is soluble so the solution stays clear	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$

In every case the metal is oxidised from 0 to +2 and hydrogen in water is reduced from +1 to 0, so these are redox reactions. The solutions formed are alkaline — hence "alkaline earth" metals.

Reaction with Oxygen

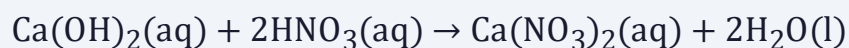
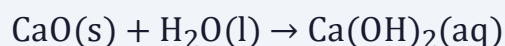
All four burn in oxygen to give the white ionic oxide:



Magnesium burns with the familiar brilliant white flame. Reactivity towards oxygen also increases down the group, and barium is stored under oil because it tarnishes so quickly in air.

Oxides and Hydroxides are Basic

Group 2 oxides react with water to give the hydroxide, and both oxides and hydroxides neutralise acids:



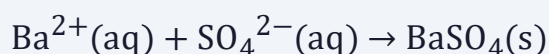
Because hydroxide solubility rises down the group, the pH of the saturated solution also rises: a saturated $\text{Mg}(\text{OH})_2$ solution is only weakly alkaline (about pH 10), while saturated $\text{Ba}(\text{OH})_2$ is strongly alkaline.

Solubility Trends

Compound type	Trend down the group	Illustration
---------------	----------------------	--------------

Hydroxides, $M(OH)_2$	Solubility increases	$Mg(OH)_2$ sparingly soluble (milk of magnesia) → $Ba(OH)_2$ soluble
Sulfates, MSO_4	Solubility decreases	$MgSO_4$ soluble (Epsom salts) → $BaSO_4$ essentially insoluble
Carbonates, MCO_3	All insoluble	$CaCO_3$ is limestone, chalk and seashell

The two opposite trends are the basis of two standard tests. Barium sulfate is so insoluble that adding acidified barium chloride solution to an unknown gives a **white precipitate** only if sulfate ions are present:



It is also why a "barium meal" is safe to swallow for an X-ray: soluble barium salts are highly toxic, but barium sulfate does not dissolve in the gut, so essentially no $Ba^{2+}(aq)$ is absorbed. The acid is added first to destroy any carbonate, which would otherwise also give a white precipitate.

Thermal Stability of the Carbonates

All Group 2 carbonates decompose on strong heating to the oxide and carbon dioxide:



Carbonate	Ionic radius of M^{2+} / nm	Approximate decomposition temperature / °C
$MgCO_3$	0.072	540
$CaCO_3$	0.100	900
$SrCO_3$	0.118	1290
$BaCO_3$	0.135	1360

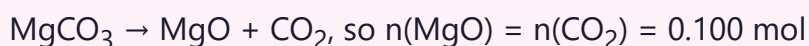
Charge density: The ratio of an ion's charge to its size. For a given charge, a smaller ion has a higher charge density and therefore a more concentrated electric field around it.

The explanation runs through polarisation. Mg^{2+} is small and 2+, so it has a high charge density. It attracts electron density from the large, easily distorted carbonate ion, weakening one of the C–O bonds and making it easier to break away as CO_2 . Going down the group the cation gets larger, its charge density falls, the carbonate ion is distorted less, and a higher temperature is needed. So **thermal stability increases down the group**. The same argument explains why Group 1 carbonates (1+ ions, lower charge density) are far more stable than Group 2 carbonates of similar size.

Worked Example — Decomposing a Carbonate:

8.43 g of magnesium carbonate is heated strongly until there is no further mass change. Calculate the mass of solid remaining and the volume of carbon dioxide produced at RTP. (M_r : $\text{MgCO}_3 = 84.3$, $\text{MgO} = 40.3$; molar gas volume = $24.0 \text{ dm}^3 \text{ mol}^{-1}$)

$$n(\text{MgCO}_3) = 8.43 / 84.3 = 0.100 \text{ mol}$$



$$\text{Mass of MgO} = 0.100 \times 40.3 = \mathbf{4.03 \text{ g}}$$

$$\text{Volume of CO}_2 = 0.100 \times 24.0 = \mathbf{2.40 \text{ dm}^3}$$

Check: mass lost = $8.43 - 4.03 = 4.40 \text{ g}$, which is 0.100 mol of CO_2 ($M_r = 44.0$). Consistent.

Where Group 2 Compounds Are Used

Compound	Use	Why that compound
Mg(OH)_2	Antacid ("milk of magnesia")	A base, but only sparingly soluble, so it neutralises stomach acid without making the stomach alkaline
CaCO_3	Indigestion tablets; flue-gas desulfurisation	Cheap, mild base that neutralises acid
Ca(OH)_2	Agricultural lime, spread on acidic soil	Soluble enough to raise soil pH quickly

BaSO_4	"Barium meal" for X-ray imaging of the gut	Opaque to X-rays and so insoluble that toxic $\text{Ba}^{2+}(\text{aq})$ is not absorbed
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	Epsom salts, used in bath salts	Very soluble, so it dissolves readily in bath water

Exam Tip: Two trends in this section run in opposite directions and are easy to mix up. Hydroxide solubility *increases* down the group; sulfate solubility *decreases*. A useful anchor is the pair of practical facts: $\text{Ca}(\text{OH})_2$ is used as limewater because it dissolves a little, and BaSO_4 is safe to swallow because it does not dissolve at all.